

ShapeDream: Zero-Shot 3D Shape Completion

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Motivation

Problem statement:

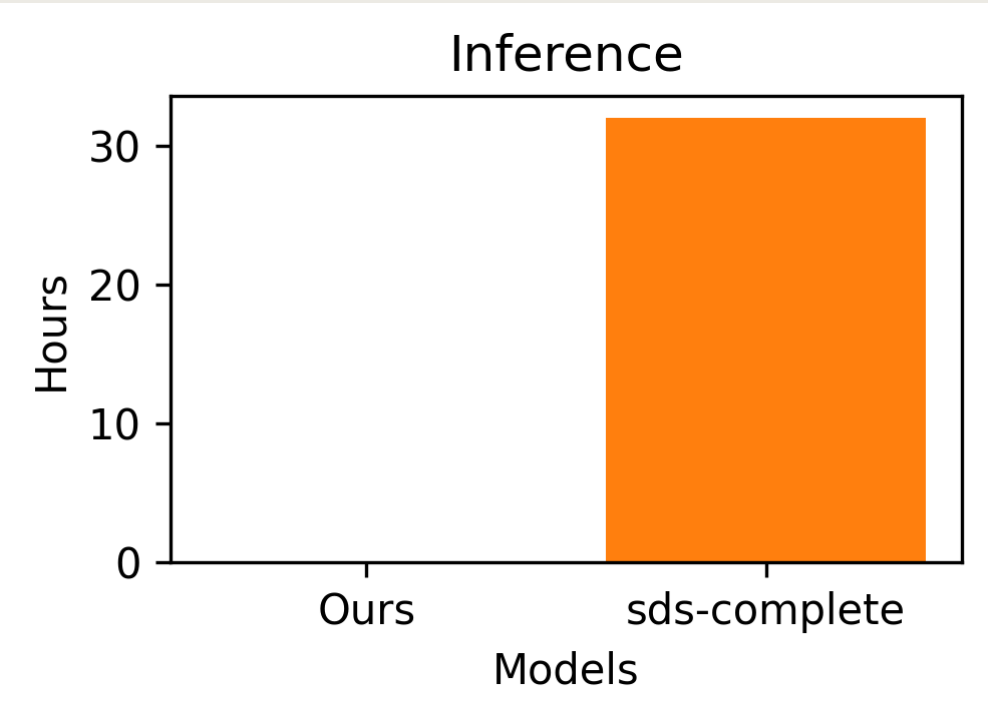
- Real-world point clouds are often incomplete

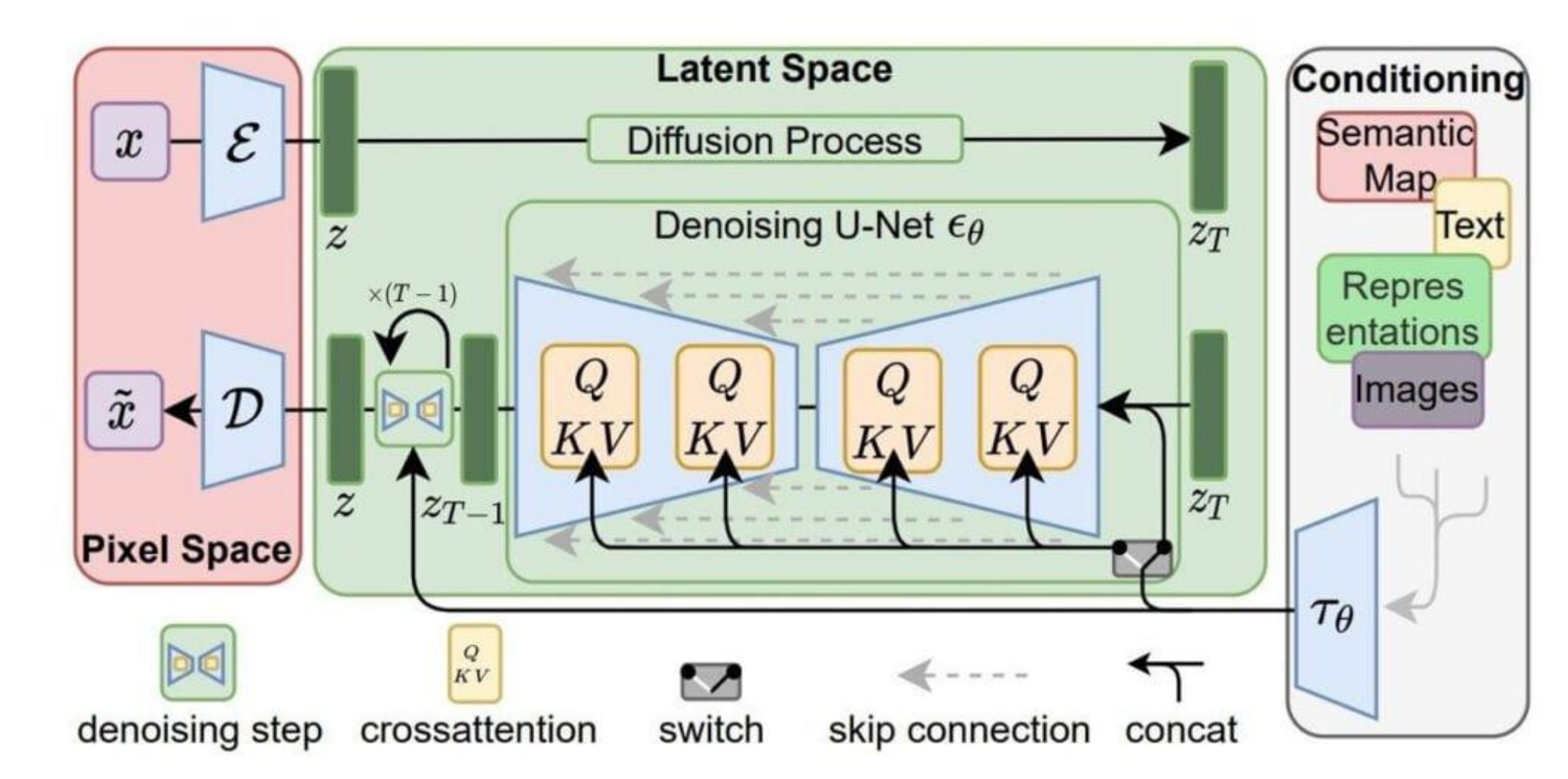
Current solutions:

- Training model from scratch
- Test time inference (e.g. SDS-complete)

Research Gap:

- No exploit of pretrained (diffusion) models
- Long inference time



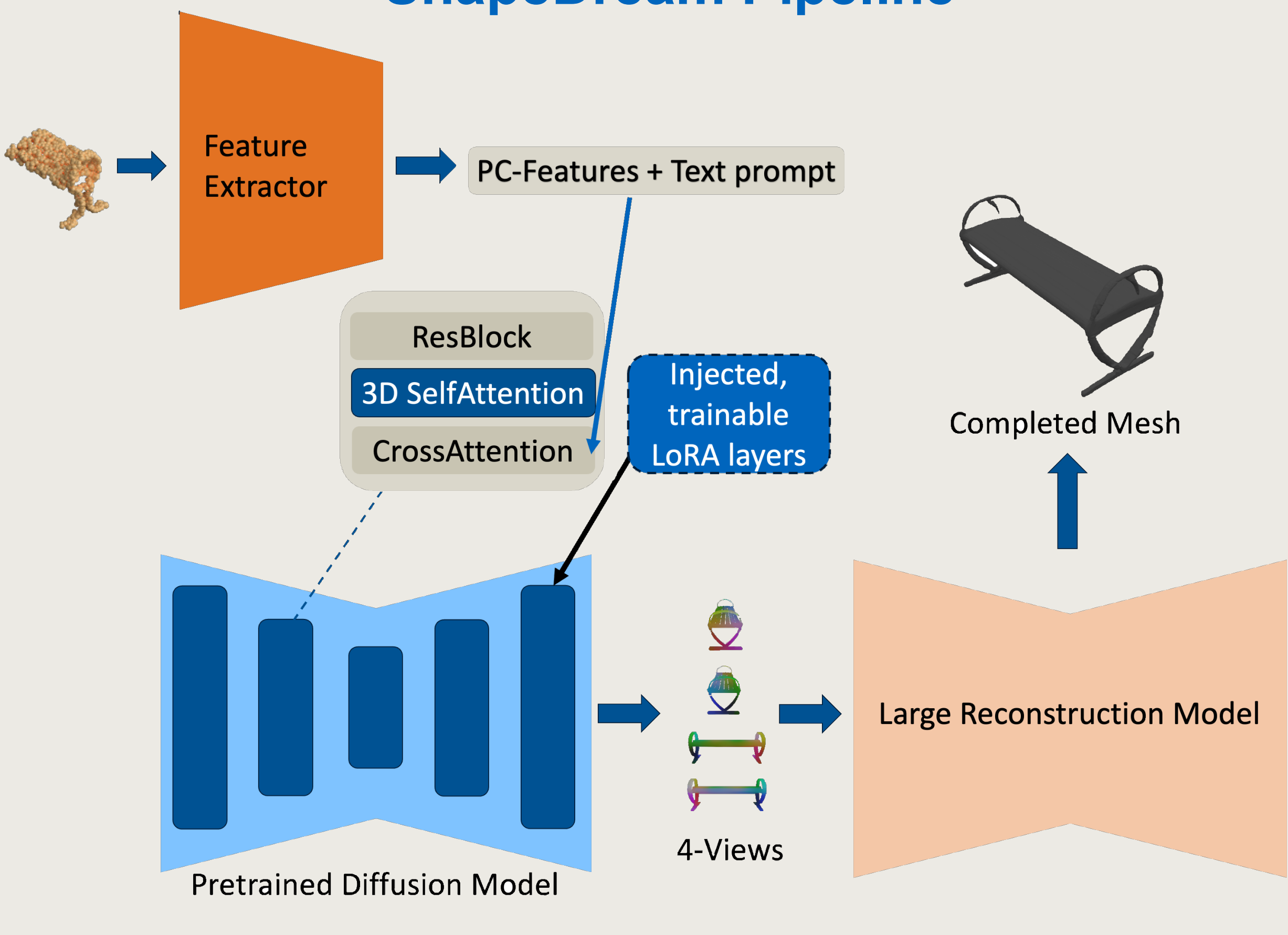


Main ideas

- Exploiting a large **pretrained** diffusion model with view consistent outputs (e.g. MVDream)
- Training only in **2D** space and using a **LRM** to generate 3D outputs
- **LoRA** finetuning using feature extractor as conditioning
- **Fast** training + **Fast** inference (inference time ~30s on an A40)
- **Competitive** Shape completions and high-quality output meshes

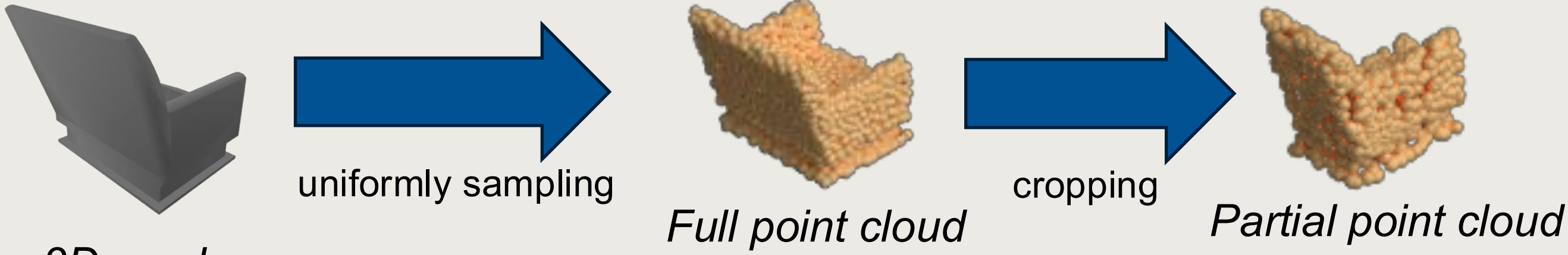


ShapeDream Pipeline

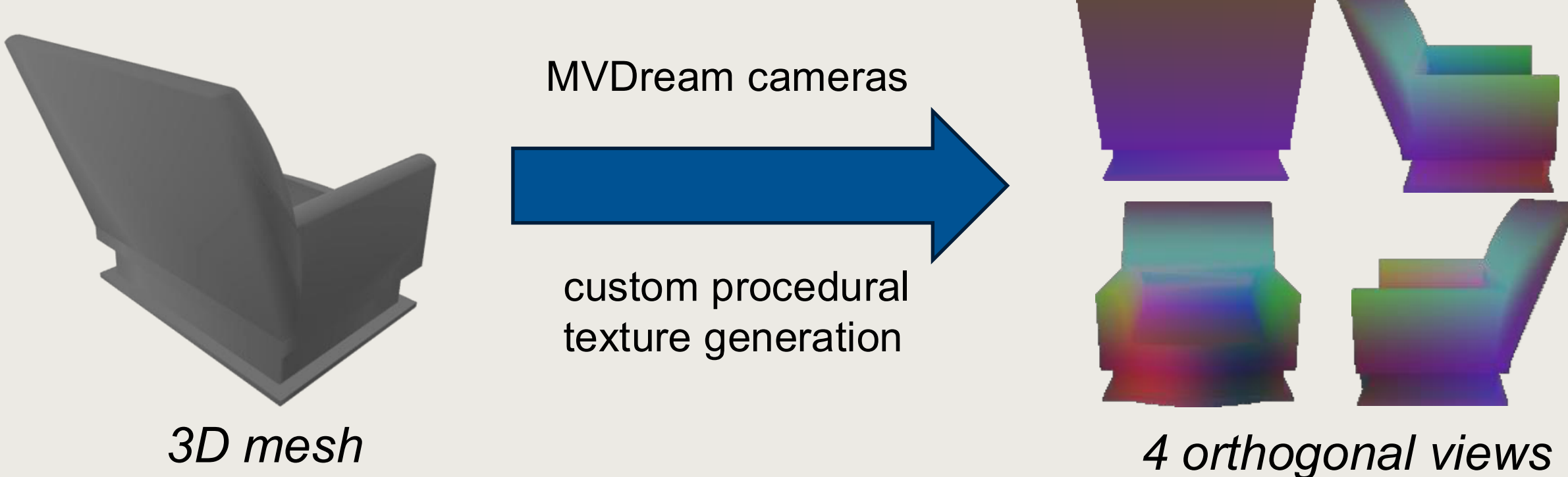


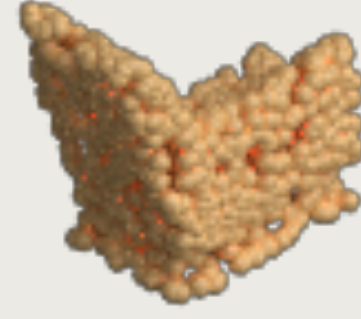
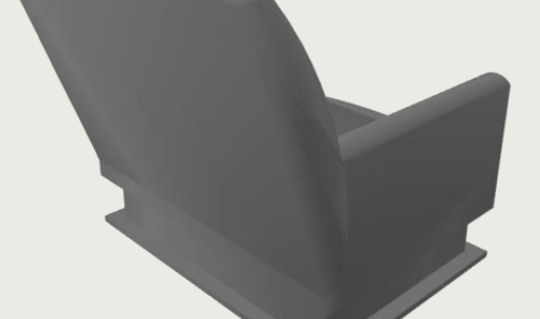
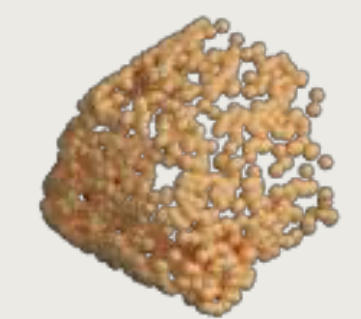
Training Dataset Generation

Point cloud generation:



Ground truth generation:



Results				
Input	Shapeformer-1	Shapeformer-2	Ours	Ground Truth
				
				
				

Category	Method	CD ↓ (10 ³)	UHD ↓ (10 ²)	F-Score ↑ (%)	TMD ↓ (10 ³)	Haus ↓ (10 ²)	<i>n</i>
Bench	ShapeFormer	16.389	26.359	23.322	128.718	26.359	181
	Ours	6.358	18.960	37.574	79.574	18.960	181
Car	ShapeFormer	15.391	24.242	20.494	130.543	24.242	181
	Ours	7.716	23.169	23.648	97.018	23.169	181
Chair	ShapeFormer	31.160	33.458	14.179	193.059	33.458	181
	Ours	17.598	26.267	21.048	140.503	26.267	181
Table	ShapeFormer	35.304	32.988	15.426	201.119	32.988	181
	Ours	27.317	28.084	17.359	173.245	28.084	181

Limitations

- Output Mesh not perfectly consistent with input
- Strong conditioning on point cloud features leads to poor out-of-domain performance

Open Future Work: Test time inference and training on more data and classes